

Gevrey class and beyond in PDE uncertainty quantification with application to random hypersurfaces

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→ *See also Andreas Rupp's talk on Wednesday at 16:10–16:35*

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Part I: Quasi-Monte Carlo methods

Quasi-Monte Carlo (QMC) methods are a class of equal weight cubature rules

$$\int_{[0,1]^s} f(\mathbf{y}) \, d\mathbf{y} \approx \frac{1}{n} \sum_{i=1}^n f(\mathbf{t}_i),$$

where $(\mathbf{t}_i)_{i=1}^n$ is an ensemble of *deterministic* nodes in $[0, 1]^s$.

The nodes $(\mathbf{t}_i)_{i=1}^n$ are NOT random!! Instead, they are *deterministically chosen*.

QMC methods exploit the smoothness and anisotropy of an integrand in order to achieve better-than-Monte Carlo rates.

Lattice rules

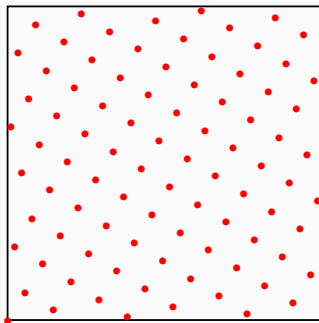
Rank-1 lattice rules

$$Q_{s,n}(f) = \frac{1}{n} \sum_{i=1}^n f(\mathbf{t}_i)$$

have the points

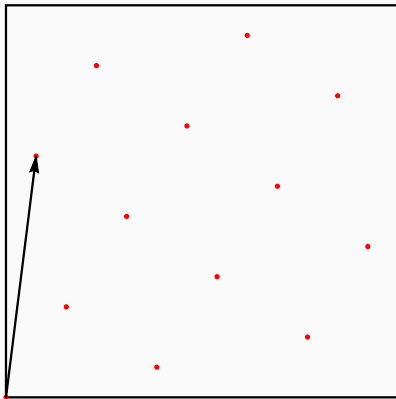
$$\mathbf{t}_i = \text{mod} \left(\frac{i\mathbf{z}}{n}, 1 \right), \quad i \in \{1, \dots, n\},$$

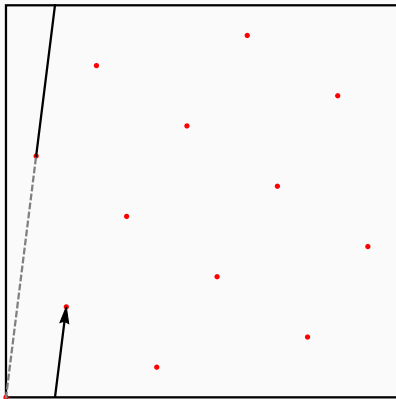
where the entire point set is determined by the *generating vector* $\mathbf{z} \in \mathbb{N}^s$, with all components *coprime* to n .

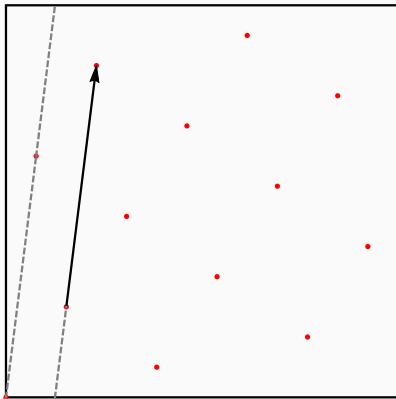


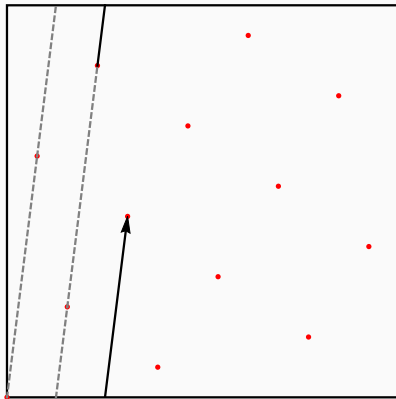
Lattice rule with $\mathbf{z} = (1, 55)$ and $n = 89$
nodes in $[0, 1]^2$

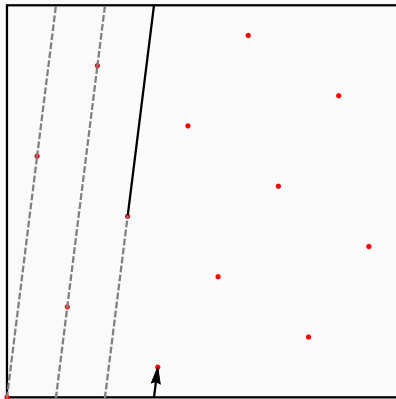
The quality of the lattice rule is determined by the choice of $\mathbf{z}$.

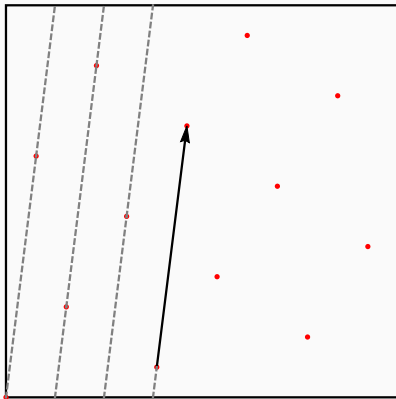


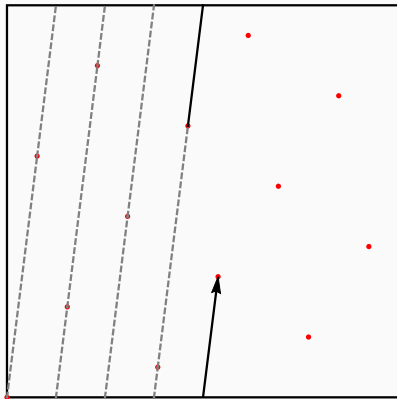


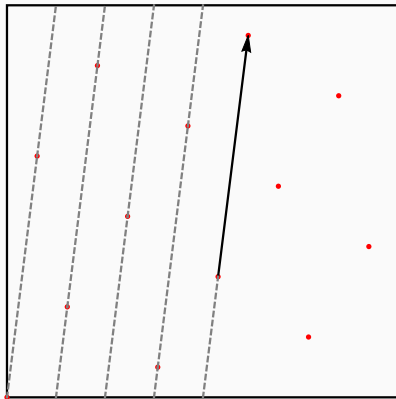


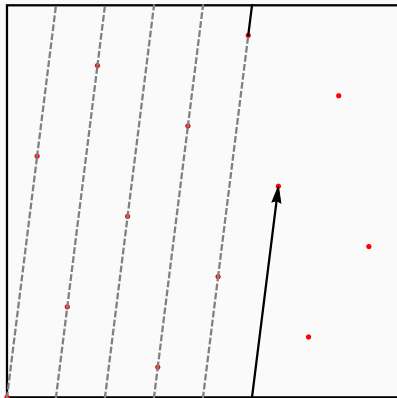


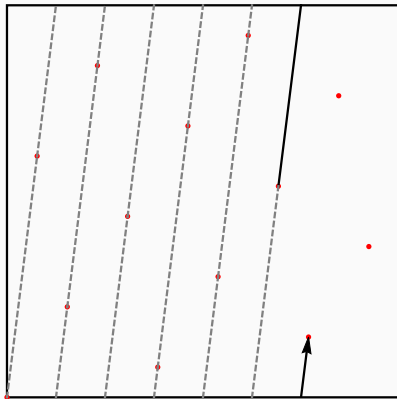


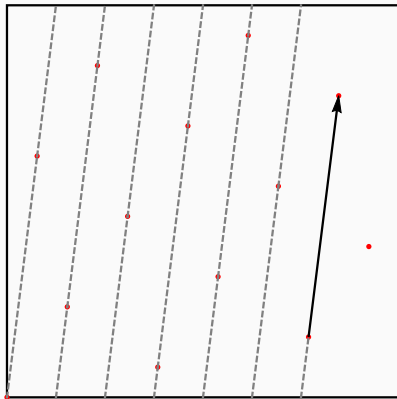


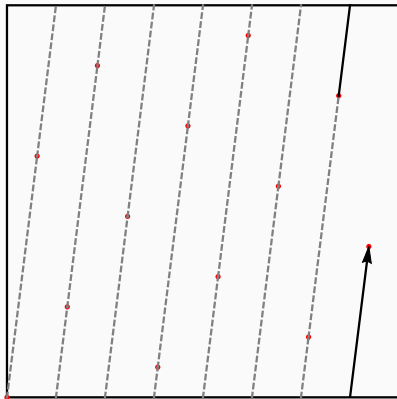












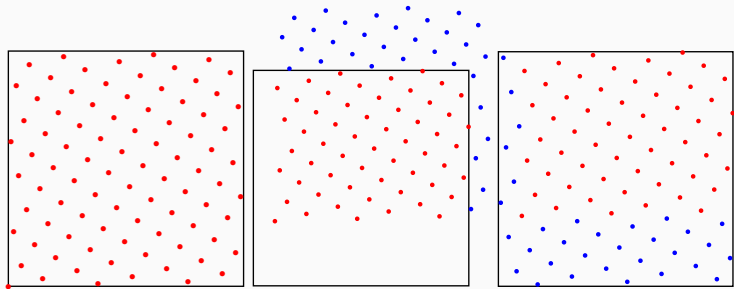
Randomly shifted lattice rules

Shifted rank-1 lattice rules have points

$$\mathbf{t}_i = \text{mod} \left(\frac{i\mathbf{z}}{n} + \mathbf{\Delta}, 1 \right), \quad i \in \{1, \dots, n\}.$$

$\mathbf{\Delta} \in [0, 1)^s$ is the **shift**

Use a number of random shifts for error estimation.



Lattice rule shifted by $\mathbf{\Delta} = (0.1, 0.3)$.

Let $\mathbf{\Delta}^{(r)}$, $r = 1, \dots, R$, be independent random shifts drawn from $U([0, 1]^s)$ and define

$$Q_{s,n}^{(r)}(f) := \frac{1}{n} \sum_{i=1}^n f(\text{mod}(\mathbf{t}_i + \mathbf{\Delta}^{(r)}, 1)). \quad (\text{QMC rule with 1 random shift})$$

Then

$$\overline{Q}_{s,n}(f) = \frac{1}{R} \sum_{r=1}^R Q_{s,n}^{(r)} f \quad (\text{QMC rule with } R \text{ random shifts})$$

is an unbiased estimator of $I_s(f)$.

Theorem (CBC error bound)

Let $n = 2^k$ be the number of lattice points. The generating vector $\mathbf{z} \in \mathbb{N}^s$ constructed by the CBC algorithm, minimizing the squared shift-averaged worst-case error $[e_{s,n,\gamma}^{\text{sh}}(\mathbf{z})]^2$ for the weighted unanchored Sobolev space in each step, satisfies

$$[e_{s,n,\gamma}^{\text{sh}}(\mathbf{z})]^2 \leq \left(\frac{2}{n} \sum_{\emptyset \neq \mathbf{u} \subseteq \{1, \dots, s\}} \gamma_{\mathbf{u}}^{\lambda} \left(\frac{2\zeta(2\lambda)}{(2\pi^2)^{\lambda}} \right)^{|\mathbf{u}|} \right)^{1/\lambda} \quad \text{for all } \lambda \in (1/2, 1],$$

where $\zeta(x) := \sum_{k=1}^{\infty} k^{-x}$ denotes the Riemann zeta function for $x > 1$.

Remarks:

- Optimal rate of convergence $\mathcal{O}(n^{-1+\varepsilon})$ in weighted Sobolev spaces, independently of s under an appropriate condition on the weights.
- Cost of algorithm for POD weights is $\mathcal{O}(s n \log n + s^2 n)$ using FFT.

Significance: Suppose that $f \in H_{s,\gamma}$ for all $\gamma = (\gamma_u)_{u \subseteq \{1, \dots, s\}}$. Then for any given sequence of weights γ , we can use the CBC algorithm to obtain a generating vector satisfying the error bound

$$\sqrt{\mathbb{E}_{\Delta} |I_s f - Q_{s,n}^{\Delta} f|^2} \leq \left(\frac{2}{n} \sum_{\emptyset \neq u \subseteq \{1, \dots, s\}} \gamma_u^{\lambda} \left(\frac{2\zeta(2\lambda)}{(2\pi^2)^{\lambda}} \right)^{|u|} \right)^{1/(2\lambda)} \|f\|_{s,\gamma} \quad (1)$$

for all $\lambda \in (1/2, 1]$. We can use the following strategy:

- For a given integrand f , estimate the norm

$$\|f\|_{s,\gamma} := \left(\sum_{u \subseteq \{1, \dots, s\}} \frac{1}{\gamma_u} \int_{[0,1]^{|u|}} \left(\int_{[0,1]^{s-|u|}} \frac{\partial^{|u|}}{\partial \mathbf{y}_u} f(\mathbf{y}) d\mathbf{y}_{-u} \right)^2 d\mathbf{y}_u \right)^{1/2}.$$

- Find weights γ which **minimize** the error bound (1).
- Using the optimized weights γ as input, use the CBC algorithm to find a generating vector which **satisfies** the error bound (1).

Part II: Parameterization of input uncertainty

Consider the elliptic PDE problem:

$$\begin{cases} -\nabla \cdot (a(\mathbf{x})\nabla u(\mathbf{x})) = f(\mathbf{x}) & \text{for } \mathbf{x} \in D, \\ +\text{boundary conditions.} \end{cases}$$

In practice, one or several of the material/system parameters may be uncertain or incompletely known and modeled as random fields:

- PDE coefficient a may be uncertain;
- Source term f may be uncertain;
- Boundary conditions may be uncertain;
- The domain D itself may be uncertain.

In **forward uncertainty quantification**, one is interested in assessing how uncertainties in the inputs of a mathematical model affect the output.

In **inverse uncertainty quantification**, one is typically interested in computing the statistics for the posterior distribution of unknown model parameters conditioned on measurements of the system response.

→ *See also Laura Bazahica's talk on Wednesday at 11:25–11:50*

Background

A popular model in the literature: the uniform and affine model.

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. For $\mathbf{x} \in D$ and $\omega \in \Omega$,

$$a(\mathbf{x}, \omega) = \bar{a}(\mathbf{x}) + \sum_{j=1}^{\infty} Y_j(\omega) \psi_j(\mathbf{x}), \quad Y_j \text{ i.i.d. uniform on } [-\frac{1}{2}, \frac{1}{2}].$$

Computing $\mathbb{E}[u(\mathbf{x}, \cdot)]$ (or some quantity of interest $\mathbb{E}[G(u)]$) using

- Rank-1 lattice cubature rules with random shifts
 $\Rightarrow$ **dimension-independent** cubature error $\mathcal{O}(n^{-1+\varepsilon})$ at best.
(Kuo, Schwab, Sloan 2012)
- Interlaced polynomial lattice rules
 $\Rightarrow$ higher order **dimension-independent** convergence $\mathcal{O}(n^{-1/p})$ for $0 < p < 1$ (p is a summability exponent s.t. $(\|\psi_j\|_{L^\infty})_{j \geq 1} \in \ell^p$).
(Dick, Kuo, Le Gia, Nuyens, Schwab 2014)

Gevrey regular random inputs

Chernov and Lê (2024a and 2024b) observed that the input random field can be much more general while retaining dimension-independent QMC convergence rates for the PDE response. It is enough that the input random field satisfies a certain parametric regularity bound.

Identifying $y_j \equiv Y_j(\omega)$ as parameters, the parametric coefficient $a: D \times [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}} \rightarrow \mathbb{R}$ is called **Gevrey regular** if it satisfies

$$\|\partial_{\mathbf{y}}^{\boldsymbol{\nu}} a(\cdot, \mathbf{y})\|_{L^\infty(D)} \leq C(|\boldsymbol{\nu}|!)^\beta \mathbf{b}^{\boldsymbol{\nu}} \quad \text{for all } \boldsymbol{\nu} \in \mathbb{N}_0^\infty, \mathbf{y} \in [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}$$

for some constant $C > 0$ and Gevrey parameter $\beta \geq 1$.

Here, we use multi-index notation

$$\partial_{\mathbf{y}}^{\boldsymbol{\nu}} = \prod_{j=1}^{\infty} \frac{\partial^{\nu_j}}{\partial y_j^{\nu_j}}, \quad \mathbf{b}^{\boldsymbol{\nu}} = \prod_{j=1}^{\infty} b_j^{\nu_j}, \quad |\boldsymbol{\nu}| = \sum_{j=1}^{\infty} \nu_j.$$

A new type of model

We can generalize the analysis for even more general models.

Let $a: D \times [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}} \rightarrow \mathbb{R}$ satisfy

$$\|\partial_{\mathbf{y}}^{\boldsymbol{\nu}} a(\cdot, \mathbf{y})\|_{L^{\infty}(D)} \leq C \sum_{\substack{\mathbf{m} \leq \boldsymbol{\nu} \\ \text{supp}(\mathbf{m}) = \text{supp}(\boldsymbol{\nu})}} (|\mathbf{m}|!)^{\beta} \prod_{j \in \text{supp}(\mathbf{m})} b_j^{m_j} \quad \text{for all } \boldsymbol{\nu} \in \mathbb{N}_0^{\infty}, \mathbf{y} \in [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}.$$

(In the special case $\boldsymbol{\nu} = \mathbf{0}$, the sum is taken to be equal to 1.)

Parametric regularity bounds of this flavor arise in certain transformations of random variables.

Example

Consider

$$a(\mathbf{x}, \mathbf{y}) = a_0(\mathbf{x}) + \sum_{j=1}^{\infty} \xi(y_j) \psi_j(\mathbf{x}), \quad y_j \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}([-1/2, 1/2]).$$

- $z_j = \xi(y_j) = \sin(2\pi y_j) \sim \text{Arcsine}(-1, 1)$
 - This is the “periodic model of UQ”, studied e.g. by [Kaarnioja–Kuo–Sloan 2020], [Kaarnioja–Kazashi–Kuo–Nobile–Sloan 2022], [Gilbert–Giles–Kuo–Sloan–Srikumar 2025+], . . .
 - Special, since lattice rules have higher-order convergence in the periodic setting.
- $z_j = \xi(y_j) = y_j^2 \sim \text{Beta}(\frac{1}{2}, 1)$ (beta distribution)
- $z_j = \xi(y_j) = y_j^k \sim \text{Power}(1/k)$ (power distribution)
- $z_j = \xi(y_j) = e^{y_j} \sim \text{LogUniform}(e^{-1/2}, e^{1/2})$ (log-uniform distribution)
- $\vdots$

Part III: Application to PDE uncertainty quantification

Modeling assumption

Let $D \subset \mathbb{R}^d$, $d \in \{1, 2, 3\}$, be a nonempty, bounded Lipschitz domain and $f \in H^{-1}(D)$ a fixed source term. Let us investigate the variational problem, for $\mathbf{y} \in [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}$, to find $u(\cdot, \mathbf{y}) \in H_0^1(D)$ such that

$$\int_D a(\mathbf{x}, \mathbf{y}) \nabla u(\mathbf{x}, \mathbf{y}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x} = \int_D f(\mathbf{x}) v(\mathbf{x}) \, d\mathbf{x} \quad \text{for all } v \in H_0^1(D),$$

where the diffusion coefficient $a: D \times [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}} \rightarrow \mathbb{R}$ is assumed to satisfy

$$\|\partial_{\mathbf{y}}^{\nu} a(\cdot, \mathbf{y})\|_{L^{\infty}(D)} \leq C \sum_{\substack{\mathbf{m} \leq \nu \\ \text{supp}(\mathbf{m}) = \text{supp}(\nu)}} (|\mathbf{m}|!)^{\beta} \prod_{j \in \text{supp}(\mathbf{m})} b_j^{m_j} \quad \text{for all } \nu \in \mathbb{N}_0^{\infty}, \mathbf{y} \in [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}.$$

In addition, we assume that:

(A1) there exists a positive constant $a_{\min}$ such that $a(\mathbf{x}, \mathbf{y}) \geq a_{\min}$ for all $\mathbf{x} \in D$ and $\mathbf{y} \in [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}$

(A2) $(b_j)_{j \geq 1} \in \ell^p$ for some $0 < p < 1$

(A3) $b_1 \geq b_2 \geq \dots \geq 0$

If we start from the assumption

$$\|\partial_{\mathbf{y}}^{\boldsymbol{\nu}} a(\cdot, \mathbf{y})\|_{L^\infty(D)} \leq C \sum_{\substack{\mathbf{m} \leq \boldsymbol{\nu} \\ \text{supp}(\mathbf{m}) = \text{supp}(\boldsymbol{\nu})}} (|\mathbf{m}|!)^\beta \prod_{j \in \text{supp}(\mathbf{m})} b_j^{m_j} \quad \text{for all } \boldsymbol{\nu} \in \mathbb{N}_0^\infty$$

and differentiate the variational formulation

$$\int_D a(\mathbf{x}, \mathbf{y}) \nabla u(\mathbf{x}, \mathbf{y}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x} = \int_D f(\mathbf{x}) v(\mathbf{x}) \, d\mathbf{x} \quad \text{for all } v \in H_0^1(D)$$

on both sides w.r.t. $\partial_{\mathbf{y}}^{\boldsymbol{\nu}}$, we ultimately obtain the recurrence

$$\begin{aligned} & \|\partial_{\mathbf{y}}^{\boldsymbol{\nu}} u(\cdot, \mathbf{y})\|_{H_0^1(D)} \\ & \leq \frac{C}{a_{\min}} \sum_{0 \neq \mathbf{m} \leq \boldsymbol{\nu}} \binom{\boldsymbol{\nu}}{\mathbf{m}} \sum_{\substack{\mathbf{w} \leq \mathbf{m} \\ \text{supp}(\mathbf{w}) = \text{supp}(\mathbf{m})}} (|\mathbf{w}|!)^\beta \left(\prod_{j \geq 1} b_j^{w_j} \right) \|\partial_{\mathbf{y}}^{\boldsymbol{\nu} - \mathbf{m}} u(\cdot, \mathbf{y})\|_{H_0^1(D)}. \end{aligned}$$

Solving this recurrence leads to the parametric regularity bound

$$\begin{aligned} & \|\partial_{\mathbf{y}}^{\boldsymbol{\nu}} u(\cdot, \mathbf{y})\|_{H_0^1(D)} \\ & \leq \frac{\|f\|_{H^{-1}(D)}}{a_{\min}} \sum_{\mathbf{m} \leq \boldsymbol{\nu}} \max\{1, C\}^{|\mathbf{m}|} 2^{|\mathbf{m}|-1} (|\mathbf{m}|!)^\beta \left(\prod_{j \geq 1} \tilde{b}_j^{m_j} S(\nu_j, m_j) \right), \end{aligned}$$

where $\tilde{b}_j = \frac{b_j}{a_{\min}}$.

Dimension dependent QMC convergence

One can use this result to deduce the existence of “good” Sobolev weights:¹

$$\gamma_u = \frac{1}{\left(\frac{2\zeta(2\lambda)}{(2\pi^2)^\lambda}\right)^{|u|/(\lambda+1)}} \sum_{\mathbf{v} \subset u} \left(\max\{1, C\}^{|\mathbf{v}|} 2^{|\mathbf{v}|-1} (|\mathbf{v}|!)^\beta \prod_{j \in \mathbf{v}} \tilde{b}_j \right)^{\frac{2}{\lambda+1}}, \quad u \subset \{1, \dots, s\},$$

where we can take

$$\lambda = \begin{cases} \frac{p}{2-p} & \text{if } p \in (\frac{2}{3}, \frac{1}{\beta}), \\ \frac{1}{2-2\varepsilon} \text{ for arbitrary } \varepsilon \in (0, 1) & \text{if } p \in (0, \min\{\frac{2}{3}, \frac{1}{\beta}\}], \quad p \neq \frac{1}{\beta}, \end{cases}$$

to ensure that

$$\sqrt{\mathbb{E}_{\Delta} \|I_s(u_s) - Q_{n,\Delta}(u_s)\|_{H_0^1(D)}^2} = \mathcal{O}(n^{\max\{-1/p+1/2, -1+\varepsilon\}}),$$

where the rate is independent of the dimension s .

Dimension-independent QMC convergence works analogously to the Gevrey setting!

¹The strategy is similar to that used in [Kaarnioja–Kazashi–Kuo–Nobile–Sloan 2022] and, recently, [Kaarnioja–Rupp–Gopalakrishnan 2025+].

Dimension truncation error

Interestingly, there seems to be one key difference to the “usual” Gevrey setting which arises within the context of the so-called **dimension truncation error**.

- If the parametric input is in the Gevrey class

$\|\partial_{\mathbf{y}}^{\nu} a(\cdot, \mathbf{y})\|_{L^{\infty}(D)} \lesssim (|\nu|!)^{\beta} \mathbf{b}^{\nu}$, then the “Taylor series argument” used by [Guth–Kaarnioja–Kuo–Schillings–Sloan 2024] yields the dimension truncation error rate

$$\left\| \int_{[-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}} u(\cdot, \mathbf{y}) \, d\mathbf{y} - \int_{[-\frac{1}{2}, \frac{1}{2}]^s} u_s(\cdot, \mathbf{y}_{\leq s}) \, d\mathbf{y}_{\leq s} \right\|_{H_0^1(D)} = \mathcal{O}(s^{-2/p+1}).$$

- However, if the parametric input satisfies

$\|\partial_{\mathbf{y}}^{\nu} a(\cdot, \mathbf{y})\|_{L^{\infty}(D)} \lesssim \sum_{\mathbf{m} \leq \nu, \text{supp}(\mathbf{m})=\text{supp}(\nu)} (|\mathbf{m}|!)^{\beta} \mathbf{b}^{\mathbf{m}}$, then the same proof technique yields the **reduced** dimension truncation error rate

$$\left\| \int_{[-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}} u(\cdot, \mathbf{y}) \, d\mathbf{y} - \int_{[-\frac{1}{2}, \frac{1}{2}]^s} u_s(\cdot, \mathbf{y}_{\leq s}) \, d\mathbf{y}_{\leq s} \right\|_{H_0^1(D)} = \mathcal{O}(s^{-1/p+1}).$$

Here $u_s(\cdot, \mathbf{y}_{\leq s}) := u(\cdot, (y_1, \dots, y_s, 0, 0, \dots))$.

Numerical experiments

Let $D = (0, 1)^2$ and consider the problem

$$\begin{cases} -\nabla \cdot (a(\mathbf{y}) \nabla u(\mathbf{x}, \mathbf{y})) = f(\mathbf{x}), & \mathbf{x} \in D, \mathbf{y} \in [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}, \\ u(\cdot, \mathbf{y})|_{\partial D} = 0, & \mathbf{y} \in [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}. \end{cases}$$

For simplicity, we assume that the diffusion coefficient $a(\mathbf{y})$ is **independent** of $\mathbf{x} \in D$ in what follows, allowing us to write

$$\mathbb{E}[u(\mathbf{x}, \cdot)] = v(\mathbf{x}) \mathbb{E}\left[\frac{1}{a}\right], \quad \text{where } v \text{ is the solution to } \begin{cases} -\Delta v = f \text{ in } D, \\ v|_{\partial D} = 0. \end{cases}$$

This means that we can focus on the integration problem

$$\mathbb{E}\left[\frac{1}{a}\right] = \int_{[-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}} \frac{d\mathbf{y}}{a(\mathbf{y})}.$$

We use the “off-the-shelf” generating vector

lattice-39101-1024-1048576.3600 (lattice D) from

<https://web.maths.unsw.edu.au/~fkuo/lattice/>

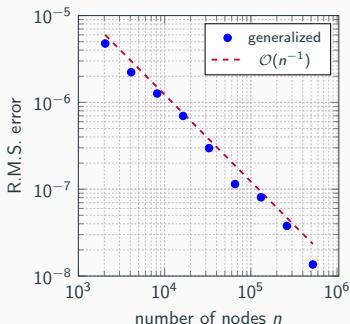
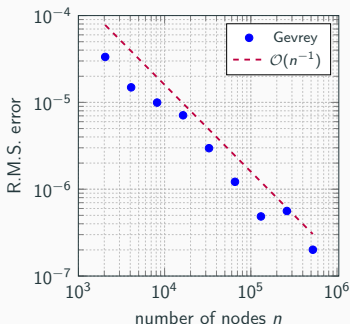
in all numerical experiments.

QMC error

We fix the truncation dimension $s = 1\,000$ and use $R = 16$ random shifts to estimate the root-mean-square (R.M.S.) error.

We compare the QMC error rates for

$$a(\mathbf{y}) = \frac{1}{1.5 + \sum_{j=1}^s j^{-1.5} y_j} \quad (\text{Gevrey}) \quad \text{vs.} \quad a(\mathbf{y}) = \frac{1}{1.5 + \sum_{j=1}^s j^{-1.5} y_j^3} \quad (\text{generalized})$$

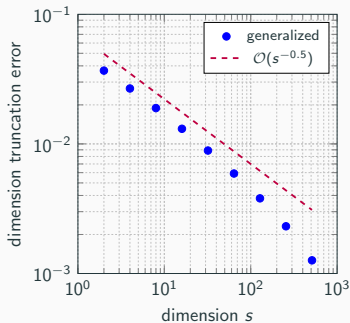
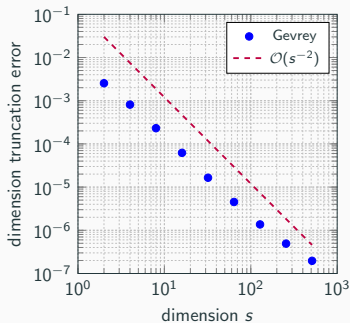


No meaningful difference in QMC performance between the smoothness classes.

Dimension truncation error

We compare the dimension truncation error rates for

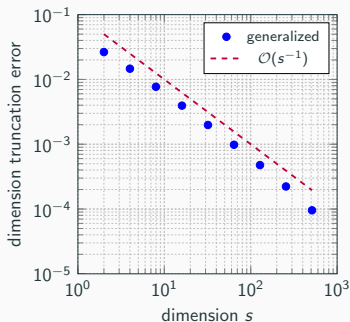
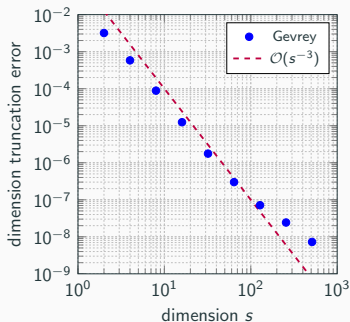
$$a(\mathbf{y}) = \frac{1}{1.5 + \sum_{j=1}^{\infty} j^{-1.5} y_j} \quad (\text{Gevrey}) \quad \text{vs.} \quad a(\mathbf{y}) = \frac{1}{1.5 + \sum_{j=1}^{\infty} j^{-1.5} y_j^2} \quad (\text{generalized})$$



Dimension truncation error

We compare the dimension truncation error rates for

$$a(\mathbf{y}) = \frac{1}{1 + \sum_{j=1}^{\infty} j^{-2} y_j} \quad (\text{Gevrey}) \quad \text{vs.} \quad a(\mathbf{y}) = \frac{1}{1 + \sum_{j=1}^{\infty} j^{-2} y_j^2} \quad (\text{generalized})$$



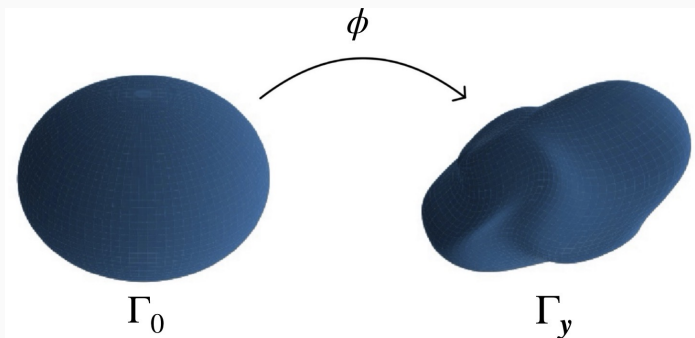
Part IV: Uncertainty quantification for surfaces

Geometric setting

Random, compact $\mathcal{C}^2$ -hypersurface $\Gamma_{\mathbf{y}} \subset \mathbb{R}^{d+1}$ without boundary, which is obtained by applying a given random field ϕ to a fixed compact reference $\mathcal{C}^2$ -hypersurface $\Gamma_0 \subset \mathbb{R}^{d+1}$, i.e.,

$$\Gamma_{\mathbf{y}} := \phi(\Gamma_0, \mathbf{y}), \quad \mathbf{y} \in U, \quad U \text{ fixed parameter set encoding randomness.}$$

→ Characterization of the random field later!



Model problem

Laplace–Beltrami equation:

$$-\Delta_{\Gamma_y} u = f \quad \text{on } \Gamma_y,$$

where $\Gamma_y = \phi(\Gamma_0, \mathbf{y})$, $\mathbf{y} \in U$, for a given random perturbation field ϕ and with the right-hand side $f \in L^2(\mathcal{D})$ being defined over the *hold-all*

$$\mathcal{D} := \bigcup_{\mathbf{y} \in U} \Gamma_y.$$

Here, the **Laplace–Beltrami operator**

$\Delta_{\Gamma} h = \nabla_{\Gamma} \cdot \nabla_{\Gamma} h$, where the

tangential gradient ∇_{Γ} of a differentiable function $h: \Gamma \rightarrow \mathbb{R}$ is

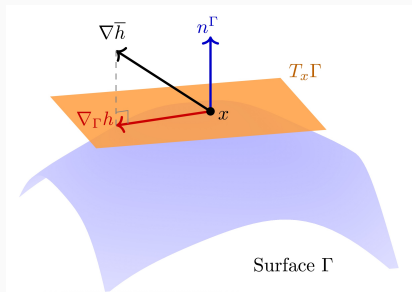
$$\nabla_{\Gamma} h := \nabla \bar{h} - \nabla \bar{h} \cdot \mathbf{n}^{\Gamma} \otimes \mathbf{n}^{\Gamma} = \pi_{\Gamma} \nabla \bar{h},$$

a projection onto the tangent space

of Γ and $\bar{h}$ denotes the smooth

extension of $h: \Gamma \rightarrow \mathbb{R}$ to a

$(d+1)$ -dimensional neighborhood V of the surface Γ such that $\bar{h}|_{\Gamma} = h$.



What is the goal?

Goal: Investigate the statistical response of the pullback PDE solution measured in terms of its expected value:

$$\mathbb{E}[u(\cdot, \mathbf{y})] = \int_U u(\cdot, \mathbf{y}) \, d\mathbf{y} \xrightarrow{\text{truncate}} \int_{U_s} u(\cdot, \mathbf{y}) \, d\mathbf{y}.$$

Problem: The stochastic dimension s can still be very high, e.g., $s = 100, 1\,000, \dots$

→ Approximate the integral using **QMC**: rank-1 lattice cubature rules with random shifts.

→ **Dimension-independent** cubature error $\mathcal{O}(n^{-1+\varepsilon})$ at best [Kuo, Schwab, Sloan 2012].

Critical assumptions

We let $U := [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}$ and $U_s := [-\frac{1}{2}, \frac{1}{2}]^s$.

(A1) For each $\mathbf{y} \in U$, let $\phi: \Gamma_0 \times U \rightarrow \Gamma_{\mathbf{y}}$ be a $\mathcal{C}^2$ -diffeomorphism such that

$$\sup_{\mathbf{y} \in U} \|\phi(\cdot, \mathbf{y})\|_{\mathcal{C}^2(\Gamma_0)} < \infty \text{ and } \sup_{\mathbf{y} \in U} \|\phi^{-1}(\cdot, \mathbf{y})\|_{\mathcal{C}^2(\Gamma_{\mathbf{y}})} < \infty$$

(A2) Let $\phi: \Gamma_0 \times U \rightarrow \Gamma_{\mathbf{y}}$ be a vector field such that, for all $\nu \in \mathbb{N}_0^\infty$, $\mathbf{y} \in U$, there holds

$$\|\partial_{\mathbf{y}}^\nu \phi(\cdot, \mathbf{y})\|_{W^{1,\infty}} \leq C \sum_{\substack{\mathbf{m} \leq \nu \\ \text{supp}(\mathbf{m}) = \text{supp}(\nu)}} (|\mathbf{m}|!)^\beta \prod_{j \in \text{supp}(\nu)} b_j^{m_j},$$

where for $\nu = \mathbf{0}$ the sum is taken to be equal to 1.

(A3) There exists a constant $C_f > 0$ and a vector $\rho = (\rho_j)_{j=1}^{d+1}$ such that

$$\|\partial_{\mathbf{x}}^\nu f\|_{L^\infty} \leq C_f \sum_{\substack{\mathbf{m} \leq \nu \\ \text{supp}(\mathbf{m}) = \text{supp}(\nu)}} (|\mathbf{m}|!)^\beta \prod_{j \in \text{supp}(\mathbf{m})} \rho_j^{m_j} \quad \text{for all } \nu \in \mathbb{N}_0^{d+1},$$

where for $\nu = \mathbf{0}$ the sum is taken to be equal to 1.

Numerical experiments

Let $\Gamma_0 := \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$ and consider the problem

$$-\Delta_{\Gamma_y} u(\cdot, \mathbf{y}) = f - \bar{f}, \quad \mathbf{x} \in \Gamma_y, \quad \mathbf{y} \in U := [-\tfrac{1}{2}, \tfrac{1}{2}]^N.$$

We model the random surface $\Gamma_y := \phi(\Gamma_0, \mathbf{y})$ using the perturbation field

$$\phi(\mathbf{x}, \mathbf{y}) = a\left(\frac{\arccos(x_3)}{\pi}, \frac{\operatorname{atan2}(x_2, x_1) + \pi}{2\pi}, \mathbf{y}\right) \mathbf{x}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \Gamma_0, \quad \mathbf{y} \in U,$$

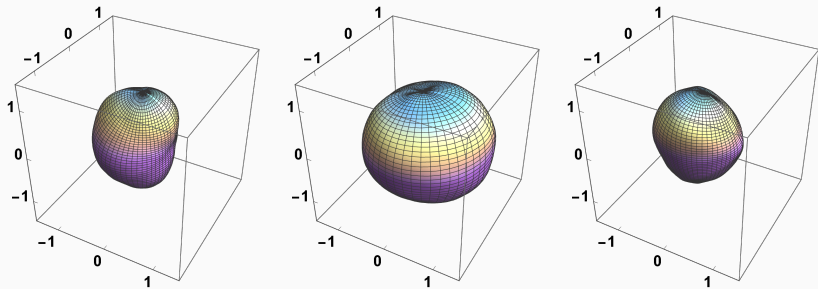
where $a((x_1, x_2), \mathbf{y}) = 1 + \sum_{j=1}^{\infty} j^{-2.1} \sin(j\pi x_1) \sin(j\pi x_2) \xi(y_j)$.

We use QMC to approximate the expected value of the pullback

$$\mathbb{E}[\hat{u}(\cdot, \mathbf{y})] = \int_U \hat{u}(\cdot, \mathbf{y}) \, d\mathbf{y}.$$

To construct the QMC rule, we use the “off-the-shelf” generating vector lattice-39101-1024-1048576.3600 (lattice D) from <https://web.maths.unsw.edu.au/~fkuo/lattice/> in all numerical experiments.

Realizations of random surfaces



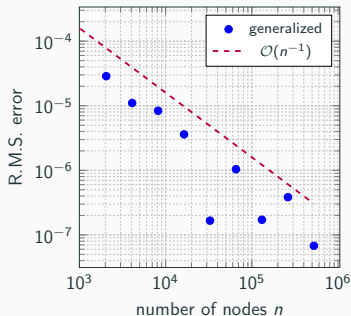
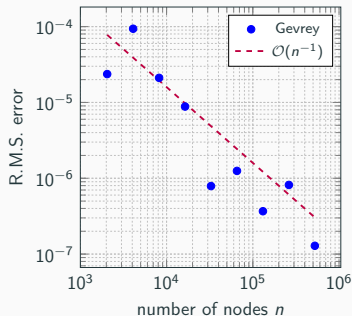
Three realizations of random surfaces corresponding to the numerical experiments with $\xi(y) = y$.

QMC error

We fix the truncation dimension $s = 100$ and use $R = 16$ random shifts to estimate the root-mean-square (R.M.S.) error.

We compare the QMC error rates for

$$\xi(y) = y \text{ (Gevrey)} \quad \text{vs.} \quad \xi(y) = y^3 \text{ (generalized)}$$



No meaningful difference in QMC performance between the smoothness classes.

Conclusions

- The generalized model enables applying QMC for random fields, where the RVs entering the representation can be transformations of uniformly distributed RVs (e.g., arcsine, beta, power distributions).
- There does not seem to be a meaningful difference in QMC performance, but the dimension truncation error rate can degenerate.
 - However, the better dimension truncation error rate may still be recovered in certain circumstances: for example, when $a(\mathbf{x}, \mathbf{y}) = a_0(\mathbf{x}) + \sum_{j=1}^{\infty} \xi(y_j) \psi_j(\mathbf{x})$ where the transformation ξ is odd. Similar phenomena have been previously reported in [Kaarnioja–Kazashi–Kuo–Nobile–Sloan 2022].

Thank you for your attention!